

5	(a)	$\frac{n_s}{n_p} = \frac{V_s}{V_p}$ $\frac{n_s}{4200} = \frac{12}{230}$ The number of turns in the secondary coil = 220	C1 A1
		Marker's comment Generally well done. A few students wrongly use the wrong formula $\frac{n_s}{n_p} = \frac{V_p}{V_s}$ or wrongly thought that one of the voltage given is the peak value.	
	(b)	(i) p.d. across the leads = e.m.f. across the output – p.d. across the lamp $V_{\text{lead}} = 12 - 11.8 = 0.2 \text{ V}$ $V_{\text{lead}} = I_{\text{lead}} \times R_{\text{lead}}$ $0.2 = I_{\text{lead}} \times 0.35$ $I_{\text{lead}} = 0.57 \text{ A}$	C1 A1
		Marker's comment Poorly done. Many did not realise that the resistance 0.35 Ω given is the resistance of the leads while the voltage 11.8 V is the p.d. across the lamp and they wrongly use them to find r.m.s. current. Another common error is that many failed to realise that the values given in the question are already the r.m.s. values and wrongly divide the values by $\sqrt{2}$ to “find” the r.m.s. values.	
		(ii) Mean power dissipated in leads $= i_{\text{rms}}^2 R = 0.571^2 \times 0.35$ $= 0.114 \text{ W}$ Maximum power $= 2P_{\text{mean}} = 2 \times 0.114$ $= 0.23 \text{ W}$	C1 A1
		Marker's comment Below satisfactory. Given that the r.m.s. current is computed in (b)(i), many did not realise or know how to use it to compute peak power, instead they compute average power. Some are carelessly use either wrong formula (e.g. $P = V^2 R$) or substitute the e.m.f. across the secondary coil into the formula.	

6	(a)	A photon is a discrete amount/quantum/packet of energy of electromagnetic radiation	P1
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